Computational Physics Lectures

Weeks 5: Entropy

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Week 5

Entropy, temperature, and the Boltzmann distribution

Abstract. This chapter develops the canonical description of equilibrium statistical mechanics from a single underlying idea: counting the microstates compatible with the information available. State counting leads first to entropy, then to temperature as an entropy versus energy slope, and then to the canonical probabilities of a small system in contact with a heat bath, namely the Boltzmann distribution. The Gibbs entropy and the maximum-entropy formulation are subsequently obtained from the same underlying counting argument.

5.1 A model heat bath

This section is concerned with the statistical behaviour of a heat bath. A heat bath is a large system that can exchange energy with a smaller observed system. Before bringing such a subsystem into contact with the bath, we first study the bath by itself as an isolated system with fixed total energy.

We use classical harmonic oscillators as a model because they are simple to visualize and their microstates can be counted analytically. The particular model is less important than the large-system behaviour that it illustrates. Our aim is to show two related properties. First, when the energy per oscillator is held fixed, the entropy becomes *extensive*: doubling both the size and the energy of a large bath approximately doubles its entropy. Second, as the bath becomes infinitely large, the curvature of its entropy as a function of energy vanishes. Over the finite range of energies that a small subsystem can exchange with the bath, the relation between the bath entropy and its energy therefore becomes linear.

This result will be used in a later section, where a small subsystem is brought into contact with the heat bath. The approximately linear bath entropy will allow us to describe the influence of the bath without following all of its microscopic degrees of freedom explicitly.

Classical harmonic oscillators

Consider N independent, identical, one-dimensional classical harmonic oscillators with Hamiltonian

$$H(\mathbf{q}, \mathbf{p}) = \sum_{i=1}^{N} \left(\frac{p_i^2}{2m} + \frac{1}{2} m\omega^2 q_i^2 \right). \tag{5.1}$$

The system is isolated, so its total energy E is fixed. The coordinates q_i and momenta p_i must attain values that collectively give this energy. Together, these values specify the available microstates. Since we have no further information that distinguishes these microstates, we make the basic statistical-mechanical assumption that all accessible microstates are equally likely. The collection of these states is the *microcanonical ensemble*.

To characterize the ensemble, we need a measure of how many microstates are compatible with its energy. In a classical system the energy is continuous, and the microstates of fixed energy form a manifold in the $2N$-dimensional phase space of the p_i and q_i. This infinitely thin energy surface ordinarily has zero phase-space volume. We therefore measure the states in a narrow energy interval from E to $E + \Delta E$.

Strictly speaking, this is not a literal count of discrete classical phase-space points. It is a dimensionless phase-space measure, expressed below in units of the cell volume h^N. A convenient way to obtain it is first to calculate the phase-space measure of all states with energy less than or equal to E; call this quantity $\Phi(E, N)$. The measure of the states in the energy shell is then, to first order in ΔE,

$$\Omega_{\Delta E}(E, N) = \Phi(E + \Delta E, N) - \Phi(E, N) \simeq \frac{\partial \Phi}{\partial E} \Delta E. \tag{5.2}$$

We seek a measure to be termed *entropy* that increases with the number of accessible microstates and is additive when independent systems are combined. The multiplicities of two independent systems multiply,

$$\Omega_{\text{tot}} = \Omega_1 \Omega_2, \tag{5.3}$$

whereas the measure for the thermodynamic entropy we seek should add. The logarithm converts this product into a sum. Apart from an overall multiplicative constant, it is the continuous function with this property. We therefore define the energy-shell entropy as

$$S_{\Delta E}(E, N) = k_{\text{B}} \ln \Omega_{\Delta E}(E, N). \tag{5.4}$$

The factor k_{B} fixes the unit of entropy. Changing the fixed resolution ΔE adds a constant to S. This convention affects the absolute entropy and some finite-size terms, but not the energy derivatives or the macroscopic conclusions considered below. For the harmonic oscillators, the enclosed state count is

$$\Phi(E, N) = \frac{1}{N!} \left(\frac{E}{\hbar\omega} \right)^N. \tag{5.5}$$

The most important feature of this expression is its dependence on energy, $\Phi \propto E^N$. Increasing the available energy enlarges the allowed region in each of the $2N$ phase-space directions, and the resulting geometrical volume grows as the Nth power of E.

A word of caution: the compact expression may look reminiscent of counting energy quanta because $E/(\hbar\omega)$ is dimensionless. This resemblance should not be interpreted literally. In the present classical derivation, $\hbar$ enters only through the choice of the phase-space cell volume h^N.

The factor $1/N!$ comes from the geometrical volume of a $2N$-dimensional ball, not from treating energy quanta as indistinguishable. A quantum calculation gives a different finite-N counting formula, although the classical expression is recovered in the appropriate large-energy limit.

If you accept the above expression for the state count, and are mainly interested in the thermodynamic consequences that follow from it, you can skip the derivation from geometrical considerations of the expression in the box below.

Optional derivation: phase-space volume inside the energy surface

Introduce scaled coordinates

$$x_i = \frac{p_i}{\sqrt{m}}, \qquad y_i = \sqrt{m}\,\omega q_i. \tag{5.6}$$

The Hamiltonian becomes

$$H = \frac{1}{2}\sum_{i=1}^{N}(x_i^2 + y_i^2). \tag{5.7}$$

Hence $H \le E$ defines a $2N$-dimensional ball of radius $R = \sqrt{2E}$. Dividing phase space into cells of volume h^N, the enclosed state count is

$$\Phi(E,N) = \frac{1}{h^N}\int_{H(\mathbf{q},\mathbf{p})\le E} \mathrm{d}^N q\,\mathrm{d}^N p. \tag{5.8}$$

The coordinate transformation has Jacobian

$$\mathrm{d}^N q\,\mathrm{d}^N p = \frac{1}{\omega^N}\,\mathrm{d}^N x\,\mathrm{d}^N y. \tag{5.9}$$

The volume of a D-dimensional ball is

$$V_D(R) = \frac{\pi^{D/2}}{\Gamma(D/2+1)}R^D, \tag{5.10}$$

where the Γ function generalizes the factorial function into non-integers. With $D = 2N$ and $R = \sqrt{2E}$,

$$V_{2N}(\sqrt{2E}) = \frac{\pi^N (2E)^N}{N!}, \tag{5.11}$$

where $\Gamma(D/2+1) = \Gamma(N+1) = N!$ has been used. The factor π^N appears because $\pi^{D/2} = \pi^N$ in $2N$ dimensions. Including the Jacobian and the phase-space cell factor gives

$$\Phi(E,N) = \frac{1}{(h\omega)^N}\frac{\pi^N (2E)^N}{N!}. \tag{5.12}$$

Finally, $h = 2\pi\hbar$ gives

$$\boxed{\Phi(E,N) = \frac{1}{N!}\left(\frac{E}{\hbar\omega}\right)^N}. \tag{5.13}$$

The number of states in the energy shell of width ΔE is now

$$\Omega_{\Delta E}(E,N) = \frac{\partial \Phi}{\partial E}\Delta E = \frac{E^{N-1}}{(N-1)!(\hbar\omega)^N}\Delta E. \tag{5.14}$$

The energy-shell entropy therefore becomes

$$S_{\Delta E}(E,N) = k_{\mathrm{B}}\left[(N-1)\ln E - \ln((N-1)!) - N\ln(\hbar\omega) + \ln\Delta E\right]. \tag{5.15}$$

Its energy derivatives are

$$\frac{\partial S}{\partial E} = \frac{(N-1)k_{\mathrm{B}}}{E}, \tag{5.16}$$

$$\frac{\partial^2 S}{\partial E^2} = -\frac{(N-1)k_{\mathrm{B}}}{E^2}. \tag{5.17}$$

Where the subscript ΔE on $S_{\Delta E}$ has been left out, as it will be in the rest of the chapter, since the derivatives are independent of ΔE.

The entropy becomes linear over a finite energy interval

We first investigate the property needed for treating the oscillators as a heat bath: how the entropy varies with energy as the bath size is increased. To compare baths of different sizes under equivalent macroscopic conditions, we evaluate the entropy derivatives at the same energy per oscillator, u:

$$u = \frac{E}{N}, \tag{5.18}$$

so that

$$E_N = Nu. \tag{5.19}$$

At fixed u, the first derivative now becomes:

$$\left.\frac{\partial S}{\partial E}\right|_{E=Nu} = \frac{k_{\mathrm{B}}}{u}\left(1 - \frac{1}{N}\right), \tag{5.20}$$

and the second derivative:

$$\left.\frac{\partial^2 S}{\partial E^2}\right|_{E=Nu} = -\frac{k_{\mathrm{B}}}{Nu^2}\left(1 - \frac{1}{N}\right). \tag{5.21}$$

Consequently,

$$\left.\frac{\partial S}{\partial E}\right|_{E=Nu} \longrightarrow \frac{k_{\mathrm{B}}}{u} \qquad \text{as } N \longrightarrow \infty, \tag{5.22}$$

while

$$\left.\frac{\partial^2 S}{\partial E^2}\right|_{E=Nu} \longrightarrow 0 \qquad \text{as } N \longrightarrow \infty. \tag{5.23}$$

The entropy slope therefore approaches a size-independent constant, whereas the curvature decreases as $1/N$. More generally, at fixed $u = E/N$, the mth energy derivative scales as

$$\frac{\partial^m S}{\partial E^m} \propto N^{1-m}. \tag{5.24}$$

Thus the slope remains finite, the curvature vanishes as $1/N$, and successive higher derivatives vanish still faster. Over a fixed energy interval, the entropy of a large system is consequently increasingly well approximated by a linear function of energy.

Extensivity at fixed energy per oscillator

The preceding calculation concerned the variation of the entropy with energy and showed that the entropy of a large bath becomes locally linear over a fixed energy interval. We now consider a different question: how the value of the entropy itself scales when both the number of oscillators and the total energy are increased while $u = E/N$ is kept fixed. At this energy, the entropy is

$$S(N) = k_{\mathrm{B}} \left[(N-1)\ln(Nu) - \ln((N-1)!) - N\ln(\hbar\omega) + \ln \Delta E \right]. \tag{5.25}$$

The following box gives the detailed large-N algebra. It may be skipped on a first reading. The result needed below is that, when E/N is fixed, the entropy per oscillator S/N approaches a constant. As the entropy scales with the system size, entropy becomes an extensive property, and can be used in macroscopic thermodynamics building on the present microscopic foundation.

Optional derivation: the macroscopic limit using Stirling's formula

For large n, Stirling's formula gives

$$n! \simeq \sqrt{2\pi n}\left(\frac{n}{e}\right)^n, \tag{5.26}$$

and hence

$$\ln(n!) \simeq n\ln n - n + \frac{1}{2}\ln(2\pi n). \tag{5.27}$$

Using $n = N-1$, we obtain

$$\ln((N-1)!) \simeq (N-1)\ln(N-1) - (N-1) + \frac{1}{2}\ln\big(2\pi(N-1)\big). \tag{5.28}$$

Inserting this into the energy-shell entropy gives

$$\begin{aligned}\frac{S(N)}{k_{\mathrm{B}}} \simeq{}& (N-1)\ln(Nu) - (N-1)\ln(N-1) + (N-1) \\ &- \frac{1}{2}\ln\big(2\pi(N-1)\big) - N\ln(\hbar\omega) + \ln\Delta E.\end{aligned} \tag{5.29}$$

The first two terms can be combined:

$$\begin{aligned}(N-1)\ln\left(\frac{Nu}{N-1}\right) &= (N-1)\ln\left(\frac{u}{1-1/N}\right) \\ &= (N-1)\ln u - (N-1)\ln\left(1-\frac{1}{N}\right).\end{aligned} \tag{5.30}$$

Since the Taylor series for $-\ln(1-x)$ is

$$-\ln(1-x) = x + \frac{x^2}{2} + \frac{x^3}{3} + \cdots, \tag{5.31}$$

we have

$$\begin{aligned}\lim_{N\to\infty} -(N-1)\ln\left(1-\frac{1}{N}\right) &= \lim_{x\to 0}\left(\frac{1}{x}-1\right)\left(x + \frac{x^2}{2} + \frac{x^3}{3} + \cdots\right) \\ &= \lim_{x\to 0}(1-x)\left(1 + \frac{x}{2} + \frac{x^2}{3} + \cdots\right) \\ &= 1.\end{aligned} \tag{5.32}$$

Consequently, for large N, Eq. (5.30) becomes

$$(N-1)\ln\left(\frac{Nu}{N-1}\right) \simeq (N-1)\ln u + 1. \tag{5.33}$$

The leading terms in the entropy are therefore proportional to N:

$$S(N) = Nk_{\mathrm{B}}\left[1 + \ln\left(\frac{u}{\hbar\omega}\right)\right] + O(\ln N). \tag{5.34}$$

The notation $O(\ln N)$ represents finite-size contributions whose magnitude grows no faster than $\ln N$. These corrections become negligible compared with the leading term, which grows as N, because

$$\frac{\ln N}{N} \longrightarrow 0 \qquad \text{as } N \longrightarrow \infty. \tag{5.35}$$

Consequently,

$$\frac{S(N)}{N} \longrightarrow k_{\mathrm{B}}\left[1 + \ln\left(\frac{u}{\hbar\omega}\right)\right]. \tag{5.36}$$

Thus, at fixed energy per oscillator $u = E/N$, each additional oscillator contributes approximately the same amount to the total entropy. This is what is meant by saying that entropy is *extensive*. If both the number of oscillators and the total energy are multiplied by the same factor λ, so that u remains unchanged, then

$$S(\lambda E, \lambda N) \simeq \lambda S(E, N). \tag{5.37}$$

For example, joining two large systems with the same energy per oscillator approximately doubles both the number of oscillators and the entropy. The relation is not exact for finite N because of the $O(\ln N)$ correction, but its relative importance vanishes as the system becomes large. The energy-shell entropy is therefore extensive in the macroscopic limit.

Conclusion: the large bath

We can now summarize the behaviour of the model heat bath. At fixed energy per oscillator, its entropy is extensive in the macroscopic limit. In addition,

$$\left.\frac{\partial^2 S}{\partial E^2}\right|_{E=Nu} \longrightarrow 0 \qquad \text{as } N \longrightarrow \infty. \tag{5.38}$$

The vanishing curvature means that the entropy slope changes negligibly when a finite amount of energy is given to or taken from an infinitely large bath.

Choose a reference energy E_0 for the bath and write $\delta E = E - E_0$. The Taylor expansion of the bath entropy is

$$S(E) = S(E_0) + \left.\frac{\partial S}{\partial E}\right|_{E_0}\delta E + \frac{1}{2}\left.\frac{\partial^2 S}{\partial E^2}\right|_{E_0}(\delta E)^2 + \cdots. \tag{5.39}$$

For an infinitely large bath, and for an exchanged energy δE that remains finite compared with

the total bath energy, the quadratic and higher-order terms vanish. Defining

$$a = \left.\frac{\partial S}{\partial E}\right|_{E_0}, \qquad b = S(E_0) - aE_0, \tag{5.40}$$

we obtain

$$\boxed{S(E) = aE + b.} \tag{5.41}$$

Here a is the constant entropy slope of the idealized infinitely large bath, and b is an offset fixed by the chosen reference state. We do not yet attach a temperature interpretation to a.

The linear relation is local in a macroscopic sense: it applies over the range of energies exchanged with a much smaller subsystem. It does not imply that the entropy of a finite bath is globally linear over energy changes that scale with the size of the bath. In a later section, the result $S(E) = aE + b$ will be used to determine how contact with the bath weights the possible energies of a small subsystem.

5.2 Two heat baths in thermal contact

The preceding section established an important property of a large heat bath. Over an energy range that is small compared with its total energy, its entropy is approximately linear, and its entropy slope changes only very little when a finite amount of energy is added or removed. We now use this result to discuss what happens when two large systems are allowed to exchange energy.

Consider two heat baths, labelled 1 and 2. Before they are brought into contact, each bath is isolated and has a fixed energy. After contact, the two baths may exchange energy, but the compound system remains isolated. Its total energy is therefore fixed:

$$E_{\text{tot}} = E_1 + E_2. \tag{5.42}$$

For a particular division of the total energy, the baths have respectively $\Omega_1(E_1)$ and $\Omega_2(E_2)$ accessible microstates. A microstate of the compound system consists of one microstate of bath 1 together with one microstate of bath 2. The number of compound-system microstates compatible with this energy division is consequently

$$\Omega_{\text{tot}}(E_1) = \Omega_1(E_1)\Omega_2(E_{\text{tot}} - E_1). \tag{5.43}$$

The accessible microstates of the isolated compound system are assumed to be equally likely. An energy division represented by more compound-system microstates will therefore be observed more frequently than one represented by fewer microstates.

Using $S = k_{\text{B}} \ln \Omega$, the product of multiplicities becomes a sum:

$$S_{\text{tot}}(E_1) = S_1(E_1) + S_2(E_{\text{tot}} - E_1). \tag{5.44}$$

Because the logarithm is increasing, the energy division that maximizes S_{tot} is also the division that maximizes Ω_{tot}.

Suppose a small amount of energy δE is transferred from bath 2 to bath 1. Then

$$\delta E_1 = \delta E, \qquad \delta E_2 = -\delta E. \tag{5.45}$$

To first order, the corresponding change in total entropy is

$$\delta S_{\text{tot}} = \left(\frac{\partial S_1}{\partial E_1} - \frac{\partial S_2}{\partial E_2} \right) \delta E. \tag{5.46}$$

If the two entropy slopes differ, one sign of δE gives $\delta S_{\text{tot}} > 0$. That direction of energy transfer leads to more compound-system microstates and is therefore statistically favoured. We can then predict the direction in which energy will flow, so the two baths are not yet in thermal equilibrium.

At thermal equilibrium, no direction of energy transfer is statistically preferred. The total entropy must therefore be stationary with respect to the division of energy:

$$\delta S_{\text{tot}} = 0, \tag{5.47}$$

which requires

$$\frac{\partial S_1}{\partial E_1} = \frac{\partial S_2}{\partial E_2}. \tag{5.48}$$

For a stable equilibrium, this stationary point is a maximum. Since $E_2 = E_{\text{tot}} - E_1$,

$$\frac{\partial^2 S_{\text{tot}}}{\partial E_1^2} = \frac{\partial^2 S_1}{\partial E_1^2} + \frac{\partial^2 S_2}{\partial E_2^2} < 0. \tag{5.49}$$

For the oscillator baths considered above, both contributions are negative, cf. Eq. (5.17). The equality of the entropy slopes is the condition that characterizes thermal equilibrium. This suggests introducing a quantity that has the same value for two systems in thermal equilibrium. We define temperature by

$$\boxed{\frac{1}{T} = \left(\frac{\partial S}{\partial E} \right)_{N,V,\ldots}}. \tag{5.50}$$

The equilibrium condition can then be written

$$T_1 = T_2. \tag{5.51}$$

The derivative is taken with particle number, volume, and any other external parameters held fixed. Temperature is therefore not introduced as an unrelated empirical parameter. It characterizes the entropy slope whose equality means that we can no longer predict a preferred direction of energy flow.

The direction of energy flow also follows immediately. If $T_1 < T_2$, then

$$\frac{1}{T_1} > \frac{1}{T_2}, \tag{5.52}$$

so transferring energy from bath 2 to bath 1 increases the total entropy. For the positive-temperature systems considered here, energy therefore flows statistically from the bath with higher temperature to the bath with lower temperature, until their entropy slopes, and hence their temperatures, are equal.

Optional derivation: the maximum is sharply defined for large baths

Near the equilibrium energy E_1^*, expand the total entropy to second order:

$$S_{\text{tot}}(E_1) \simeq S_{\text{tot}}(E_1^*) + \frac{1}{2} \left.\frac{\partial^2 S_{\text{tot}}}{\partial E_1^2}\right|_{E_1^*} (E_1 - E_1^*)^2. \tag{5.53}$$

The linear term vanishes because the entropy is stationary at equilibrium. The probability of a particular division of energy is proportional to the number of compatible compound-system microstates:

$$P(E_1) \propto \Omega_{\text{tot}}(E_1) = \exp\left(\frac{S_{\text{tot}}(E_1)}{k_{\text{B}}}\right). \tag{5.54}$$

Inserting the expansion gives an approximately Gaussian distribution around E_1^*:

$$\begin{aligned} P(E_1) &\propto \exp\left(\frac{S_{\text{tot}}(E_1^*)}{k_{\text{B}}}\right) \exp\left[\frac{1}{2k_{\text{B}}} \left.\frac{\partial^2 S_{\text{tot}}}{\partial E_1^2}\right|_{E_1^*} (E_1 - E_1^*)^2\right] \\ &\propto \exp\left[\frac{1}{2k_{\text{B}}} \left.\frac{\partial^2 S_{\text{tot}}}{\partial E_1^2}\right|_{E_1^*} (E_1 - E_1^*)^2\right] \end{aligned} \tag{5.55}$$

Because the second derivative is negative, departures from the equilibrium energy division are less probable. For macroscopic baths, the distribution is very narrow relative to the total energies, so the equilibrium energy division is observed overwhelmingly often.

5.3 A small system in contact with a heat bath: counting microstates of heat bath

The preceding sections established the two ingredients needed here. First, the accessible microstates of an isolated compound system are taken to be equally likely. Second, over the finite energy range exchanged with a small system, a large bath has an entropy whose slope is effectively constant. We can therefore derive the probabilities of the small system directly by counting how many bath microstates are compatible with each small-system microstate.

We now replace bath 1 by a much smaller observed system and retain bath 2 as a large heat bath. The compound system, consisting of the observed system and the bath together, is isolated. Its total energy is fixed:

$$E_{\text{tot}} = E_{\text{s}} + E_{\text{b}}. \tag{5.56}$$

The observed system may exchange energy with the bath, so its energy is not fixed. We wish to determine the probabilities of its possible microstates.

Let μ denote one microstate of the observed system, with energy E_μ. If the observed system occupies this microstate, the bath must have energy

$$E_{\text{b}} = E_{\text{tot}} - E_\mu. \tag{5.57}$$

For this one specified system microstate, the number of compatible microstates of the compound system is simply the number of bath microstates at the remaining energy:

$$\Omega_{\text{compound}}(\mu) = \Omega_{\text{b}}(E_{\text{tot}} - E_\mu). \tag{5.58}$$

All accessible microstates of the isolated compound system are equally likely. The probability of observing system microstate μ is therefore proportional to the number of bath microstates compatible with it:

$$p_\mu \propto \Omega_{\text{b}}(E_{\text{tot}} - E_\mu). \tag{5.59}$$

This is the central step. The microstates of the small system are not equally likely, even though the microstates of the isolated compound system are. A system microstate is more probable when it leaves the bath with more compatible microstates.

Using $S_{\text{b}} = k_{\text{B}} \ln \Omega_{\text{b}}$ gives

$$p_\mu \propto \exp\left[\frac{S_{\text{b}}(E_{\text{tot}} - E_\mu)}{k_{\text{B}}}\right]. \tag{5.60}$$

From the model-bath discussion, the entropy of a sufficiently large bath is approximately linear over the finite energy range exchanged with the small system. Let U be the mean energy of the observed system and define the corresponding bath energy

$$E_{\text{b}}^* = E_{\text{tot}} - U. \tag{5.61}$$

For a system microstate of energy E_μ, the bath energy can then be written as

$$E_{\text{tot}} - E_\mu = E_{\text{b}}^* - (E_\mu - U). \tag{5.62}$$

Expanding around E_{b}^* gives

$$\begin{aligned} S_{\text{b}}(E_{\text{tot}} - E_\mu) \simeq S_{\text{b}}(E_{\text{b}}^*) &- \left.\frac{\partial S_{\text{b}}}{\partial E_{\text{b}}}\right|_{E_{\text{b}}^*} (E_\mu - U) \\ &+ \frac{1}{2}\left.\frac{\partial^2 S_{\text{b}}}{\partial E_{\text{b}}^2}\right|_{E_{\text{b}}^*} (E_\mu - U)^2 + \cdots . \end{aligned} \tag{5.63}$$

For an ideal infinitely large bath, the curvature and higher derivatives vanish over this finite energy range. And since:

$$\frac{\partial S_{\text{b}}}{\partial E_{\text{b}}} = \frac{1}{T}, \tag{5.64}$$

we obtain

$$S_{\text{b}}(E_{\text{tot}} - E_\mu) \simeq S_{\text{b}}(E_{\text{b}}^*) - \frac{E_\mu - U}{T}. \tag{5.65}$$

The terms $S_{\text{b}}(E_{\text{b}}^*) + U/T$ are independent of μ and are absorbed into the normalization of the probabilities. It follows that

$$p_\mu \propto \exp\left(-\frac{E_\mu}{k_{\text{B}}T}\right). \tag{5.66}$$

Normalization introduces the partition function

$$Z = \sum_\mu \exp\left(-\frac{E_\mu}{k_{\text{B}}T}\right), \tag{5.67}$$

and the equilibrium probability becomes

$$\boxed{p_\mu = \frac{1}{Z}\exp\left(-\frac{E_\mu}{k_\mathrm{B}T}\right).} \tag{5.68}$$

This is the Boltzmann distribution.

The bath has not literally remained in the same microstate, and its entropy has not remained exactly constant. When the observed system gains energy E_μ, the bath loses that energy and its entropy changes by approximately $-E_\mu/T$. What remains unchanged in the ideal-bath limit is the bath's *entropy slope*, $\partial S_\mathrm{b}/\partial E_\mathrm{b} = 1/T$, and therefore its temperature. A *finite* bath has a small but nonzero curvature, so its temperature changes slightly and the Boltzmann distribution receives finite-bath corrections.

5.4 From state-counting entropy to Gibbs entropy

So far, we have used $S = k_\mathrm{B}\ln\Omega$, which applies directly when all accessible microstates are equally likely. In the preceding section, however, we found that a small system in contact with a heat bath generally has a nonuniform distribution of microstate probabilities. A microstate is more probable when it is compatible with a larger number of bath microstates. To continue using an entropy description, we therefore need an expression for the entropy of a general probability distribution $\{p_\mu\}$. We will derive this expression from the same state-counting argument used above. In the following section, we will use it to maximize the entropy of the small system subject to an average-energy constraint, and we will show why this is equivalent, for an ideal heat bath, to maximizing the entropy of the isolated compound system.

Imagine preparing the system in the same way and observing its microstate independently M times. Suppose that microstate μ is observed n_μ times, where

$$\sum_\mu n_\mu = M, \qquad p_\mu = \frac{n_\mu}{M}. \tag{5.69}$$

For large M, the observed fraction n_μ/M approaches the probability p_μ. We now ask how many different sequences of M observations have these same occurrence numbers $\{n_\mu\}$, and hence conform to the same empirical probability distribution $\{p_\mu\}$.

If all M entries in a sequence were different, there would be $M!$ possible orderings. However, the label μ occurs n_μ times. Permuting these identical occurrences among themselves does not produce a new sequence. We must therefore divide by $n_\mu!$ for every microstate μ. The number of distinct sequences with the specified occurrence numbers is consequently

$$W = \frac{M!}{\prod_\mu n_\mu!}. \tag{5.70}$$

For example, suppose that the system has two microstates with probabilities

$$p_1 = \frac{1}{3}, \qquad p_2 = \frac{2}{3}. \tag{5.71}$$

In a sequence of $M = 6$ observations, these probabilities correspond to observing microstate 1 twice and microstate 2 four times:

$$n_1 = 2, \qquad n_2 = 4. \tag{5.72}$$

Possible sequences include

$$(1, 1, 2, 2, 2, 2), \qquad (1, 2, 1, 2, 2, 2), \qquad (2, 1, 2, 2, 1, 2),$$

and

$$(2, 2, 1, 2, 2, 1).$$

All these sequences contain two observations of microstate 1 and four observations of microstate 2, although the observations occur in different orders. Altogether, the number of such sequences is

$$W = \frac{6!}{2!\,4!} = 15. \tag{5.73}$$

For independent observations, every sequence with these fixed occurrence numbers has the same probability, namely

$$p_1^{n_1} p_2^{n_2}. \tag{5.74}$$

Thus, the sequences conforming to a specified set of probabilities are treated on an equal footing in the counting. Notice that we are not claiming that sequences with different occurrence numbers are equally likely. Rather, for fixed occurrence numbers $\{n_\mu\}$, all orderings have the same probability, and W counts how many such orderings realize the distribution $p_\mu = n_\mu/M$.

The number W depends on the probabilities through $n_\mu = M p_\mu$. A relatively even distribution can be realized by many different sequences, whereas a distribution concentrated in one microstate can be realized by comparatively few. This dependence of the number of sequences on $\{p_\mu\}$ is what leads to the entropy of a general probability distribution.

Applying the state-counting entropy $k_\mathrm{B} \ln W$ to the complete sequence and using Stirling's approximation gives

$$\begin{aligned} \ln W &= \ln(M!) - \sum_\mu \ln(n_\mu!) \\ &\simeq M \ln M - M - \sum_\mu \left(n_\mu \ln n_\mu - n_\mu\right). \end{aligned} \tag{5.75}$$

Because $\sum_\mu n_\mu = M$, the terms linear in M and n_μ cancel:

$$\begin{aligned} \ln W &\simeq M \ln M - \sum_\mu n_\mu \ln n_\mu \\ &= M \ln M - M \sum_\mu p_\mu \ln(M p_\mu) \\ &= -M \sum_\mu p_\mu \ln p_\mu. \end{aligned} \tag{5.76}$$

The logarithm of the number of conforming sequences is therefore proportional to the number M of observations. Dividing by M gives the exponential rate at which the number of sequences compatible with the empirical distribution grows:

$$\boxed{S[p] = -k_\mathrm{B} \sum_\mu p_\mu \ln p_\mu.} \tag{5.77}$$

This is the Gibbs entropy. The counting argument shows how the logarithmic state-counting measure extends from a uniform ensemble to a nonuniform probability distribution: $-\sum_\mu p_\mu \ln p_\mu$ measures, per independent preparation, the growth in the number of sequences compatible with the empirical distribution.

If precisely Ω states are accessible and equally likely, then $p_\mu = 1/\Omega$ and

$$S[p] = -k_{\mathrm{B}} \sum_{\mu=1}^{\Omega} \frac{1}{\Omega} \ln \frac{1}{\Omega} = k_{\mathrm{B}} \ln \Omega. \tag{5.78}$$

Thus $S = k_{\mathrm{B}} \ln \Omega$ is the uniform-distribution special case of the Gibbs entropy.

There is also a useful decomposition for the isolated compound system. If subsystem microstate μ is compatible with $\Omega_{\mathrm{b},\mu} = \Omega_{\mathrm{b}}(E_{\mathrm{tot}} - E_\mu)$ bath microstates, then each compound microstate in this group has probability $p_\mu/\Omega_{\mathrm{b},\mu}$. Grouping the compound microstates according to μ therefore gives

$$\begin{aligned} S_{\mathrm{compound}}[p] &= -k_{\mathrm{B}} \sum_{\mu} \sum_{\nu=1}^{\Omega_{\mathrm{b},\mu}} \frac{p_\mu}{\Omega_{\mathrm{b},\mu}} \ln\left(\frac{p_\mu}{\Omega_{\mathrm{b},\mu}}\right) \\ &= -k_{\mathrm{B}} \sum_{\mu} p_\mu \ln p_\mu + \sum_{\mu} p_\mu k_{\mathrm{B}} \ln \Omega_{\mathrm{b},\mu}. \end{aligned}$$

Hence

$$\boxed{S_{\mathrm{compound}}[p] = S_{\mathrm{s}}[p] + \sum_{\mu} p_\mu S_{\mathrm{b}}(E_{\mathrm{tot}} - E_\mu).} \tag{5.79}$$

The first term measures how probability is distributed among subsystem microstates; the second is the average entropy of the compatible bath states. This identity is the bridge from direct counting to maximum entropy.

5.5 A small system in contact with a heat bath: maximizing entropy of subsystem

The direct counting argument above already gave the Boltzmann distribution. We now express the same physics as a constrained maximum-entropy problem involving only the observed system. This reformulation is useful computationally and conceptually, but it can appear as though maximization of the entropy of the compound system has suddenly been replaced by maximization of the subsystem entropy. The following argument explains why that replacement is justified.

Why is only the subsystem entropy maximized?

The fundamental equilibrium statement still concerns the isolated compound system. From the decomposition obtained above,

$$S_{\mathrm{compound}}[p] = S_{\mathrm{s}}[p] + \sum_{\mu} p_\mu S_{\mathrm{b}}(E_{\mathrm{tot}} - E_\mu). \tag{5.80}$$

The mean energy of the subsystem is

$$U = \sum_{\mu} p_{\mu} E_{\mu}. \tag{5.81}$$

For a large bath, its entropy is approximately linear over the range of energies exchanged with the subsystem. We may therefore write

$$S_{\mathrm{b}}(E_{\mathrm{tot}} - E_{\mu}) \simeq S_{\mathrm{b}}^{0} - aE_{\mu}, \tag{5.82}$$

where a is the essentially constant entropy slope of the bath and S_{b}^{0} is independent of μ. Therefore

$$\begin{aligned} S_{\mathrm{compound}}[p] &\simeq S_{\mathrm{s}}[p] + \sum_{\mu} p_{\mu}(S_{\mathrm{b}}^{0} - aE_{\mu}) \\ &= S_{\mathrm{s}}[p] + S_{\mathrm{b}}^{0} - aU, \end{aligned} \tag{5.83}$$

where normalization, $\sum_{\mu} p_{\mu} = 1$, has also been used. When the mean energy U is prescribed, the bath contribution $S_{\mathrm{b}}^{0} - aU$ is fixed. Maximizing the entropy of the compound system is then equivalent to maximizing $S_{\mathrm{s}}[p]$.
The bath has not been discarded. It fixes the allowed mean energy and, through its entropy slope, the energy multiplier. To see the limitation of this argument, consider a *finite* bath. Define the bath energy corresponding to the mean subsystem energy by

$$E_{\mathrm{b}}^{*} = E_{\mathrm{tot}} - U. \tag{5.84}$$

Expanding the bath entropy around this energy gives

$$\begin{aligned} S_{\mathrm{b}}(E_{\mathrm{tot}} - E_{\mu}) &= S_{\mathrm{b}}\big(E_{\mathrm{b}}^{*} - (E_{\mu} - U)\big) \\ &\simeq S_{\mathrm{b}}(E_{\mathrm{b}}^{*}) - a(E_{\mu} - U) + \frac{1}{2} S_{\mathrm{b}}''(E_{\mathrm{b}}^{*})(E_{\mu} - U)^{2} + \cdots, \end{aligned} \tag{5.85}$$

where

$$a = \left.\frac{\partial S_{\mathrm{b}}}{\partial E_{\mathrm{b}}}\right|_{E_{\mathrm{b}}^{*}}. \tag{5.86}$$

After averaging over the subsystem distribution, the linear term vanishes because $\sum_{\mu} p_{\mu}(E_{\mu} - U) = 0$. The bath contribution nevertheless retains a dependence on the energy variance through the quadratic term, and on higher moments through the remaining terms. Fixing only U therefore no longer makes the bath contribution constant. The usual subsystem maximum-entropy argument is exact within the ideal-bath approximation, in which the bath entropy is affine over the relevant subsystem energy range and the system–bath interaction energy can be neglected in the energy bookkeeping. It becomes increasingly accurate as the bath curvature vanishes.

We now perform this reduced maximization explicitly. Let the observed system have probabilities

p_μ and average energy

$$U = \sum_\mu p_\mu E_\mu. \tag{5.87}$$

As established in the preceding section, its Gibbs entropy is

$$S_\mathrm{s}[p] = -k_\mathrm{B} \sum_\mu p_\mu \ln p_\mu. \tag{5.88}$$

We therefore maximize $S_\mathrm{s}[p]$ subject to

$$\sum_\mu p_\mu = 1, \qquad \sum_\mu p_\mu E_\mu = U. \tag{5.89}$$

Introduce multipliers α and λ and define

$$\mathcal{L}[p] = -k_\mathrm{B} \sum_\mu p_\mu \ln p_\mu - \alpha \left(\sum_\mu p_\mu - 1 \right) - \lambda \left(\sum_\mu p_\mu E_\mu - U \right). \tag{5.90}$$

Stationarity with respect to each p_μ gives

$$\frac{\partial \mathcal{L}}{\partial p_\mu} = -k_\mathrm{B}(\ln p_\mu + 1) - \alpha - \lambda E_\mu = 0. \tag{5.91}$$

Hence

$$p_\mu = \exp\left(-1 - \frac{\alpha}{k_\mathrm{B}}\right) \exp\left(-\frac{\lambda E_\mu}{k_\mathrm{B}}\right). \tag{5.92}$$

Normalization gives

$$p_\mu = \frac{1}{Z(\lambda)} \exp\left(-\frac{\lambda E_\mu}{k_\mathrm{B}}\right), \tag{5.93}$$

where

$$Z(\lambda) = \sum_\mu \exp\left(-\frac{\lambda E_\mu}{k_\mathrm{B}}\right). \tag{5.94}$$

The normalization and energy constraints determine the distribution up to the multiplier λ. The identification of this multiplier is shown in the optional box below.

Optional derivation: identifying the energy multiplier

To identify λ, insert the maximizing distribution into the system entropy. Since

$$\ln p_\mu = -\frac{\lambda E_\mu}{k_\mathrm{B}} - \ln Z, \tag{5.95}$$

we obtain

$$\begin{aligned} S_\mathrm{s} &= -k_\mathrm{B} \sum_\mu p_\mu \ln p_\mu \\ &= \lambda \sum_\mu p_\mu E_\mu + k_\mathrm{B} \ln Z \sum_\mu p_\mu \\ &= \lambda U + k_\mathrm{B} \ln Z. \end{aligned} \tag{5.96}$$

Differentiating along this family of maximizing distributions gives

$$\frac{dS_{\mathrm{s}}}{dU} = \lambda. \tag{5.97}$$

At equilibrium, the entropy slope of the system equals that of the bath, so

$$\lambda = \frac{dS_{\mathrm{s}}}{dU} = \frac{\partial S_{\mathrm{b}}}{\partial E_{\mathrm{b}}} = \frac{1}{T}. \tag{5.98}$$

The maximizing distribution is therefore

$$p_\mu = \frac{1}{Z} \exp\left(-\frac{E_\mu}{k_{\mathrm{B}} T}\right). \tag{5.99}$$

Thus the constrained maximum-entropy calculation and the direct microcanonical counting argument give the same Boltzmann distribution.

Optional extension: a classical phase-space distribution

For a classical observed system, label a microstate by the phase-space point $(\mathbf{q}, \mathbf{p})$ and write its Hamiltonian as $H(\mathbf{q}, \mathbf{p})$. It is convenient to introduce the dimensionless phase-space measure

$$\mathrm{d}\Gamma = \frac{1}{h^{N_{\mathrm{s}}}} \mathrm{d}^{N_{\mathrm{s}}} q \, \mathrm{d}^{N_{\mathrm{s}}} p. \tag{5.100}$$

With this convention $P(\mathbf{q}, \mathbf{p})$ is dimensionless, and the sums above become

$$1 = \int P(\mathbf{q}, \mathbf{p}) \, \mathrm{d}\Gamma, \tag{5.101}$$

$$U = \int P(\mathbf{q}, \mathbf{p}) H(\mathbf{q}, \mathbf{p}) \, \mathrm{d}\Gamma, \tag{5.102}$$

and

$$S[P] = -k_{\mathrm{B}} \int P(\mathbf{q}, \mathbf{p}) \ln P(\mathbf{q}, \mathbf{p}) \, \mathrm{d}\Gamma. \tag{5.103}$$

The same constrained variation gives

$$P(\mathbf{q}, \mathbf{p}) = \frac{1}{Z} \exp\left[-\frac{H(\mathbf{q}, \mathbf{p})}{k_{\mathrm{B}} T}\right], \tag{5.104}$$

with

$$Z = \frac{1}{h^{N_{\mathrm{s}}}} \int \exp\left[-\frac{H(\mathbf{q}, \mathbf{p})}{k_{\mathrm{B}} T}\right] \mathrm{d}^{N_{\mathrm{s}}} q \, \mathrm{d}^{N_{\mathrm{s}}} p. \tag{5.105}$$

5.6 Summary

The argument may be summarized as follows:

1. The compound system is isolated, so its accessible microstates are taken to be equally likely.
2. Two macroscopic systems exchange energy until the total multiplicity, and therefore the total entropy, is maximal.
3. At that maximum their entropy slopes are equal. Temperature is defined through $1/T = \partial S/\partial E$.
4. A small-system microstate is weighted by the number of bath microstates compatible with the remaining energy.
5. The entropy of a very large bath is linear over the relevant energy interval, with an essentially unchanged slope $1/T$.
6. The resulting small-system probability is the Boltzmann distribution.
7. Counting sequences of many independent preparations gives the Gibbs entropy $S[p] = -k_\mathrm{B} \sum_\mu p_\mu \ln p_\mu$, which reduces to $k_\mathrm{B} \ln \Omega$ for equally likely states.
8. At prescribed average energy, maximizing the small system's Gibbs entropy reproduces the same distribution; the bath is represented by the energy constraint and its multiplier $1/T$.